

Fixed-multiplicity shuffles and balanced Smirnov words

The $e^{-(k-1)}$ law, Poisson adjacency statistics,
and explicit all-orders corrections

A research note on OEIS A007060, A193624, A330266, and related sequences

1 October 2026

Abstract

Let $B_{n,k}$ be the number of permutations of kn distinct cards, partitioned into n ranks of size k , in which neighboring cards never have the same rank. The OEIS entry A330266 [1] records the fixed- k conjecture (S. Kirgizov, 29 September 2023)

$$\frac{B_{n,k}}{(kn)!} \longrightarrow e^{-(k-1)}.$$

The case $k = 4$ was subsequently deduced on that entry from an approximation to a related sequence (D. Radcliffe, 9 September 2025), but the general statement remained posed there. We give a direct proof for every fixed $k \geq 2$ and prove considerably more. If $X_{n,k}$ is the number of equal-rank adjacencies in a uniformly random shuffle, then

$$X_{n,k} \xrightarrow{d} \text{Poisson}(k-1).$$

We derive an exact probability-generating integral whose kernel is an associated Laguerre polynomial, an exact inclusion–exclusion formula, a gamma-expectation representation, and a constructive expansion to arbitrary order. In particular, uniformly for fixed k ,

$$\begin{aligned} \frac{B_{n,k}}{(kn)!} = e^{-(k-1)} & \left[1 - \frac{(k-1)^2}{2kn} + \frac{(k-1)^2(3k^2 - 14k + 7)}{24k^2n^2} \right. \\ & \left. - \frac{(k-1)^4(k^2 - 10k + 17)}{48k^3n^3} + O_k(n^{-4}) \right]. \end{aligned}$$

The first four factorial cumulants are obtained in exact rational form; the mean is exactly $k-1$ and the variance is $(k-1) - (k-1)^2/(kn-1)$. For A330266 this gives

$$\frac{B_{n,4}}{(4n)!} = e^{-3} \left(1 - \frac{9}{8n} - \frac{3}{128n^2} + \frac{189}{1024n^3} + O(n^{-4}) \right).$$

We also treat the corresponding multiset sequences, local probabilities, Hamiltonian paths in complete multipartite graphs, ratio asymptotics, and a Lambert- W inverse transseries. Exact reproducibility code accompanies the article.

Scope and priority. The sequence definitions and historical comments below are taken from the OEIS pages cited in the bibliography. A330266 [1] now states that its particular $k = 4$ leading equivalent follows from an approximation to A190830. The point of the present note is the uniform fixed- k theorem, the Poisson law, exact factorial cumulants, and systematic correction terms. No claim of a complete literature search or historical priority is made; the results should be independently checked before submission or attribution.

Editorial note (1 October 2026, batch 73O2). Manuscript 57 of batch 73 of ProveIt’s incoming-reports intake, placed in commit `6e193dd4f`; AI-assisted, unrefereed, not formalized. The write added the label prefix `bsw:`, citation commands, the names of the OEIS contributors, notes marked “[Added 1 October 2026, batch 73O2]”, Remark 6.3, one bibliography entry and Appendix C. The analytic tail step of Theorem 6.2 is a sketch; the fixed- k limit law does not depend on it. [2 October 2026, batch 77P5.] Part II, from manuscript 11 of batch 77, proves that step for fixed k (Remark II.6.3); Sections 1–15 and Appendices A–C form Part I.

Contents

1	The OEIS problem and the family of sequences	4
2	A weighted run transform	5
3	The exact Laguerre integral	6
4	Exact inclusion–exclusion and a compact reciprocal polynomial	7
5	The Poisson adjacency theorem	8
6	A gamma expectation and an all-orders expansion engine	9
7	Exact factorial cumulants	11
8	Explicit expansion through third order	12
9	Specializations to the OEIS rows	13
10	Asymptotics of the counts, ratios, and inverse functions	14
11	Relation to rook polynomials and recent multipartite work	15
12	Reproducibility and a formalization route	16
13	Proposed OEIS updates	16
14	Further research questions and topics	17
15	Conclusion	18
A	Cumulant algebra	18
B	Exact initial terms for $k = 5$	19
C	Provenance and editorial notes	19

II	A rigorous uniform remainder for balanced Smirnov asymptotics	21
II.1	Notation of Part II	21
II.2	Scope and the gap being repaired	22
II.3	Exact model and factorial-moment domination	23
II.4	The explicit uniform operator remainder	24
II.5	Integer powers and the normalized logarithm	25
II.6	The all-orders factorial-cumulant degree argument	26
II.7	Verified formulas and consequences	28
II.8	Reproducibility and review	29
II.9	Provenance of Part II	30
II.A	Direct verification of the weighted gamma tail	31

1 The OEIS problem and the family of sequences

Fix integers $n \geq 0$ and $k \geq 2$. There are n ranks, each containing k distinct cards. A shuffle is a permutation of all $N = kn$ cards. Let

$$X_{n,k} = \#\{1 \leq j < N : \text{the cards at positions } j \text{ and } j+1 \text{ have the same rank}\}. \quad (1)$$

Write

$$B_{n,k} = N! \mathbb{P}(X_{n,k} = 0), \quad p_{n,k} = \frac{B_{n,k}}{N!}. \quad (2)$$

If cards within one rank are made indistinguishable, let $S_{n,k}$ denote the number of admissible multiset words. Every rank word lifts to exactly $(k!)^n$ labeled permutations, so

$$B_{n,k} = (k!)^n S_{n,k}, \quad p_{n,k} = \frac{S_{n,k}}{(kn)!/(k!)^n}. \quad (3)$$

Thus the same probability governs the labeled and multiset versions.

The most visible OEIS rows are as follows.

k	labeled count $B_{n,k}$	interpretation	multiset count $S_{n,k}$
2	A007060 [2]	couples with no adjacent spouses	A114938 [4]
3	A193624 [3]	triples with no adjacent siblings	A193638 [5]
4	A330266 [1]	four cards in each rank	A321633 [6]

A330266 [1] records the exact integral

$$B_{n,4} = \int_0^\infty (t^4 - 12t^3 + 36t^2 - 24t)^n e^{-t} dt$$

and the conjectured fixed-multiplicity law, posed there by S. Kirgizov on 29 September 2023,

$$p_{n,k} \longrightarrow e^{-(k-1)}. \quad (4)$$

The entry also records a September 2025 deduction of the $k = 4$ equivalent from a separate approximation: D. Radcliffe (9 September 2025) combined A330266(n) = $24^n n!$ A190830(n) with V. Kotěšovec's approximation of A190830 and Stirling's formula. Our derivation below is direct, applies to every fixed k , and identifies the full limiting distribution behind (4).

1.1 Why a Poisson law is plausible but not automatic

For each of the $N - 1$ adjacent position pairs, the probability of equal ranks is $(k - 1)/(N - 1)$. Hence

$$\mathbb{E}X_{n,k} = (N - 1) \frac{k - 1}{N - 1} = k - 1. \quad (5)$$

If the rare adjacency indicators were independent, one would expect a Poisson limit of mean $k - 1$ and hence a zero probability $e^{-(k-1)}$. They are not independent: overlapping position pairs and the fixed rank multiplicities produce both local and global dependence. The proof therefore requires more than the heuristic “rare events with bounded mean.” The exact generating function will make the cancellation transparent.

1.2 Hamiltonian-path interpretation

Replace the deck by the complete n -partite graph with n parts of size k . A vertex ordering is an admissible shuffle precisely when consecutive vertices are joined. Consequently $B_{n,k}$ is the number of *oriented* Hamiltonian paths, counted as ordered vertex lists, in $K_{k,k,\dots,k}$. Unoriented paths are counted by $B_{n,k}/2$ when $kn > 1$. This connection places the problem among balanced Smirnov words and complete multipartite Hamiltonian enumeration [7, 9].

2 A weighted run transform

It is useful to enumerate all shuffles while recording the number of bad adjacencies. Define

$$A_{n,k}(u) = \sum_{\pi \in \mathfrak{S}_{kn}} u^{X_{n,k}(\pi)}, \quad G_{n,k}(u) = \frac{A_{n,k}(u)}{(kn)!} = \mathbb{E} u^{X_{n,k}}. \quad (6)$$

Thus $A_{n,k}(0) = B_{n,k}$ and $G_{n,k}(0) = p_{n,k}$.

Theorem 2.1 (Run-transform generating function). *For indeterminates $x_1, \dots, x_n$,*

$$\sum_w u^{X(w)} x_1^{|w|_1} \dots x_n^{|w|_n} = \frac{1}{1 - \sum_{i=1}^n \frac{x_i}{1 + (1-u)x_i}}, \quad (7)$$

where the sum is over all finite words on $\{1, \dots, n\}$, including the empty word. Therefore

$$A_{n,k}(u) = (k!)^n [x_1^k \dots x_n^k] \frac{1}{1 - \sum_{i=1}^n \frac{x_i}{1 + (1-u)x_i}}. \quad (8)$$

Proof. Compress every maximal run of equal letters to one letter. The compressed word is a Smirnov word, meaning that consecutive letters are distinct. A run of letter i and length $r \geq 1$ contributes $u^{r-1} x_i^r$, whose generating series is

$$y_i = \sum_{r \geq 1} u^{r-1} x_i^r = \frac{x_i}{1 - u x_i}.$$

The classical multivariate generating function of Smirnov words [12, 13] is

$$\frac{1}{1 - \sum_i y_i / (1 + y_i)}.$$

Indeed, it follows from the ordinary word series $1/(1 - \sum_i z_i)$ by replacing each nonempty run by one marked letter, or by the standard run-decomposition bijection. Substitution gives

$$\frac{y_i}{1 + y_i} = \frac{x_i}{1 + (1-u)x_i},$$

which proves (7). Extracting k occurrences of each rank gives multiset words; labeling the k copies independently in each rank contributes $(k!)^n$ and proves (8). $\square$

3 The exact Laguerre integral

Set $v = u - 1$ and define

$$P_{k,u}(t) = k! [x^k] \exp\left(\frac{tx}{1-vx}\right). \quad (9)$$

Theorem 3.1 (Exact integral and polynomial kernel). *For every $n \geq 0$, $k \geq 2$, and complex u ,*

$$A_{n,k}(u) = \int_0^\infty e^{-t} P_{k,u}(t)^n dt. \quad (10)$$

The kernel has the explicit forms

$$P_{k,u}(t) = \sum_{j=1}^k \binom{k-1}{j-1} \frac{k!}{j!} t^j (u-1)^{k-j}, \quad (11)$$

$$= k!(u-1)^k L_k^{(-1)}\left(-\frac{t}{u-1}\right), \quad (12)$$

where the value at $u = 1$ is interpreted by continuity.

Proof. Use the coefficientwise Laplace identity

$$\frac{1}{1-s} = \int_0^\infty e^{-t} e^{ts} dt.$$

In (8), the exponential factors into n identical univariate series. Extracting x_i^k from each factor gives (10). Expanding $\exp(tx/(1-vx))$ first by the exponential degree and then extracting x^k gives (11). Finally, the standard expansion [14]

$$L_k^{(-1)}(z) = \sum_{j=1}^k \binom{k-1}{j-1} \frac{(-z)^j}{j!}$$

gives (12). □

For no equal adjacencies, $u = 0$ and $v = -1$. Write

$$P_k(t) := P_{k,0}(t) = (-1)^k k! L_k^{(-1)}(t). \quad (13)$$

The first kernels are

$$\begin{aligned} P_2(t) &= t^2 - 2t, \\ P_3(t) &= t^3 - 6t^2 + 6t, \\ P_4(t) &= t^4 - 12t^3 + 36t^2 - 24t, \\ P_5(t) &= t^5 - 20t^4 + 120t^3 - 240t^2 + 120t. \end{aligned}$$

The third line recovers the integral printed on A330266, while the theorem explains it as one row of a uniform Laguerre family.

4 Exact inclusion–exclusion and a compact reciprocal polynomial

For $0 \leq m \leq k-1$, put

$$a_{k,m} = \binom{k-1}{m} \frac{k!}{(k-m)!} = m! \binom{k}{m} \binom{k-1}{m}, \quad (14)$$

and define the degree- $(k-1)$ polynomial

$$R_k(z) = \sum_{m=0}^{k-1} a_{k,m} z^m. \quad (15)$$

The no-adjacency reciprocal polynomial is

$$C_k(z) = z^k P_k(1/z) = R_k(-z). \quad (16)$$

Theorem 4.1 (Exact finite formula). *Let $N = kn$. Then*

$$B_{n,k} = \sum_{m_1, \dots, m_n=0}^{k-1} (-1)^M (N-M)! \prod_{i=1}^n a_{k,m_i}, \quad M = m_1 + \dots + m_n, \quad (17)$$

$$= \sum_{M=0}^{n(k-1)} (N-M)! [z^M] C_k(z)^n. \quad (18)$$

More generally, the factorial probability generating function

$$\Phi_{n,k}(v) := \mathbb{E}(1+v)^{X_{n,k}} = G_{n,k}(1+v) \quad (19)$$

satisfies the exact identity

$$\Phi_{n,k}(v) = \sum_{M=0}^{n(k-1)} \frac{[z^M] R_k(z)^n}{(N)_{\underline{M}}} v^M. \quad (20)$$

Proof. From (11),

$$z^k P_{k,u}(1/z) = \sum_{m=0}^{k-1} a_{k,m} v^m z^m = R_k(vz).$$

Raise to the n th power and integrate termwise in (10). Since $\int_0^\infty t^{N-M} e^{-t} dt = (N-M)!$, division by $N!$ yields (20). Setting $v = -1$ gives (17) and (18). $\square$

Formula (18) is a rook-polynomial form of inclusion–exclusion. It uses only a degree- $(k-1)$ polynomial raised to the n th power and is efficient for exact integer computation. In particular, for $k = 4$ it reproduces

$$B_{13,4} = 3668033946384704437729512814619767610579526911188666362431432294400,$$

the standard-deck value recorded on A330266.

5 The Poisson adjacency theorem

Recall that the r th falling factorial is $(x)_{\underline{r}} = x(x-1)\cdots(x-r+1)$.

Theorem 5.1 (Factorial moments). *For fixed $k \geq 2$ and fixed $r \geq 0$,*

$$\mathbb{E}(X_{n,k})_{\underline{r}} = (k-1)^r + O_{k,r}(n^{-1}). \quad (21)$$

Consequently,

$$\boxed{X_{n,k} \xrightarrow[n \rightarrow \infty]{d} \text{Poisson}(k-1).} \quad (22)$$

Proof. The factorial moment is $r!$ times the coefficient of v^r in (20). Since every term $a_{k,m}z^m$ in $R_k(z)$ has the same z -degree as the corresponding v -degree, the denominator for this coefficient is exactly $(N)_{\underline{r}}$. The leading contribution to $[z^r]R_k(z)^n$ selects $a_{k,1}z$ from r distinct factors and 1 from the rest:

$$[z^r]R_k(z)^n = \binom{n}{r} a_{k,1}^r + O_{k,r}(n^{r-1}).$$

Here $a_{k,1} = k(k-1)$. Therefore

$$\begin{aligned} \mathbb{E}(X_{n,k})_{\underline{r}} &= \frac{r! \binom{n}{r} [k(k-1)]^r + O_{k,r}(n^{r-1})}{(kn)_{\underline{r}}} \\ &= (k-1)^r + O_{k,r}(n^{-1}). \end{aligned}$$

These are the factorial moments of $\text{Poisson}(k-1)$. Ordinary moments are finite linear combinations of factorial moments, so every ordinary moment converges. The Poisson distribution is moment-determinate; the method of moments gives (22). $\square$

Corollary 5.2 (The fixed-multiplicity OEIS law). *For every fixed integer $k \geq 2$,*

$$\boxed{\frac{B_{n,k}}{(kn)!} = p_{n,k} \longrightarrow e^{-(k-1)}.} \quad (23)$$

Equivalently,

$$S_{n,k} \sim \frac{(kn)!}{(k!)^n} e^{-(k-1)}. \quad (24)$$

Proof. Because the variables are integer-valued,

$$\mathbb{P}(X_{n,k} = 0) = \mathbb{P}(-\tfrac{1}{2} < X_{n,k} < \tfrac{1}{2}).$$

The endpoints $\pm \frac{1}{2}$ carry no mass under the limiting Poisson law, so weak convergence implies convergence of this probability. Hence $\mathbb{P}(X_{n,k} = 0) \rightarrow \mathbb{P}(\text{Poisson}(k-1) = 0) = e^{-(k-1)}$. Equation (24) follows from (3). $\square$

Remark 5.3. The proof is short because the correct object is not merely the zero-event count. The entire adjacency statistic has a stable Poisson limit. The constant $k-1$ is already forced by the exact mean (5).

6 A gamma expectation and an all-orders expansion engine

The exact finite sum admits a useful probabilistic transform. Let T_N have the gamma density

$$f_N(t) = \frac{t^N e^{-t}}{N!}, \quad t > 0, \quad T_N \sim \text{Gamma}(N+1, 1). \quad (25)$$

Since

$$\mathbb{E} T_N^{-M} = \frac{(N-M)!}{N!} = \frac{1}{(N)_{\underline{M}}} \quad (0 \leq M \leq N), \quad (26)$$

we obtain the following exact formula.

Proposition 6.1 (Gamma representation). *For $N = kn$,*

$$\boxed{\Phi_{n,k}(v) = \mathbb{E} \left[R_k \left(\frac{v}{T_N} \right)^n \right]}. \quad (27)$$

The gamma variable is concentrated at scale $\sqrt{N}$ around $N+1$. Since $R_k(w) = 1 + k(k-1)w + O_k(w^2)$, the exponent in (27) tends to $(k-1)v$. This already supplies another proof of the Poisson limit. More importantly, it supports a complete Laplace expansion.

For a nonnegative integer M , define polynomials $H_r(M)$ by

$$\prod_{j=0}^{M-1} \frac{1}{1-jx} = \sum_{r \geq 0} H_r(M) x^r. \quad (28)$$

Thus $H_r(M) = h_r(0, 1, \dots, M-1)$, a complete homogeneous symmetric polynomial. The first four are

$$H_0(M) = 1, \quad (29)$$

$$H_1(M) = \frac{M(M-1)}{2}, \quad (30)$$

$$H_2(M) = \frac{M(M-1)(3M^2 + M - 2)}{24}, \quad (31)$$

$$H_3(M) = \frac{M^2(M-1)^2(M+1)(M+2)}{48}. \quad (32)$$

Let $D = v d/dv$.

Theorem 6.2 (Constructive all-orders expansion). *Fix $k \geq 2$. As $n \rightarrow \infty$, uniformly for v in every compact subset of $\mathbb{C}$, $\Phi_{n,k}(v)$ has a Poincare expansion in integer powers of $1/n$. A coefficientwise construction is*

$$\boxed{\Phi_{n,k}(v) \sim \sum_{r \geq 0} \frac{1}{N^r} H_r(D) \left[R_k \left(\frac{v}{N} \right)^n \right], \quad N = kn.} \quad (33)$$

After expanding the bracket in $1/N$, collecting total powers gives any desired correction term. In particular, every coefficient is an explicit polynomial in v and a rational function of k , multiplied by $e^{(k-1)v}$.

Proof. Write

$$R_k(z)^n = \sum_{M=0}^{n(k-1)} r_{n,M} z^M.$$

Then (20) is

$$\Phi_{n,k}(v) = \sum_M r_{n,M} \frac{v^M}{N^M} \frac{N^M}{(N)_{\underline{M}}}.$$

For fixed M ,

$$\frac{N^M}{(N)_{\underline{M}}} = \prod_{j=0}^{M-1} \left(1 - \frac{j}{N}\right)^{-1} \sim \sum_{r \geq 0} \frac{H_r(M)}{N^r}.$$

Since D acts on v^M by multiplication by M , this is precisely the formal operator rule (33).

It remains to justify the asymptotic use when M is not fixed. The exact gamma representation (27) does so. Split the integral into $|T_N - N| \leq N^{2/3}$ and its complement. On the upper tail, gamma Chernoff bounds are enough. On the lower tail one combines the polynomial bound $|R_k(v/t)| \leq C_{k,V}(1 + t^{1-k})$ for $|v| \leq V$ with Stirling's formula; the exact negative moments show that terms whose total polynomial degree is proportional to N are in fact superexponentially small. Thus the weighted complement is smaller than every fixed power of N^{-1} . On the central interval, T_N^{-1} , $\log R_k(v/T_N)$, and the gamma density all possess Taylor expansions with remainders uniform for v in a prescribed compact set. After the change $T_N = N + \sqrt{N}y$, each coefficient is a polynomial in y times a Gaussian majorant, so termwise integration is valid to arbitrary fixed order. Odd half-powers cancel in the integrated expansion, leaving integer powers of $1/N$. The coefficientwise operator computation and the Laplace computation agree because both expand the same exact negative moments (26). This proves the Poincare expansion and the stated uniformity. $\square$

Remark 6.3 (Status of the analytic step). [Added 1 October 2026, batch 73O2.] The tail estimate is sketched. The coefficientwise operator rule is exact algebra, but the passage to an asymptotic statement uniform for v in compact sets is not proved in detail: the gamma Chernoff bounds, the superexponential smallness of terms of total degree proportional to N , and the cancellation of odd half-powers are asserted, not derived. A uniform factorial-moment bound of the type proved in the companion path-forest report [a189281-path-forest-expansions](#) of this collection (its Lemma 4.1, “Uniform factorial-moment bound”, label `spf:lem:moment-bound`, which bounds the k th factorial moment by $(\theta e^2)^k/k!$ for its own model) would make the truncation rigorous; that lemma is not proved here for the present model.

Accordingly, Theorem 6.2 and everything that uses it rest on this sketched step: the scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$ stated after Theorem 7.1, Theorem 8.1 (through “the fifth and higher factorial cumulants begin at $O_k(n^{-4})$ ”), Corollaries 8.2 and 8.3, the specializations (44)–(47), the count, ratio and inverse expansions (49), (52) and (54), and the correction terms in the proposed OEIS comments. They do not depend on it: Theorems 2.1, 3.1, 4.1, 5.1 and 7.1, Proposition 6.1, and Corollary 5.2, the fixed- k limit law that settles the conjecture on A330266. The intake's numerical check is consistent with the displayed coefficients: for $k = 2, \dots, 5$ the residual after the n^{-3} term of (42), multiplied by $e^{k-1}n^4$, is nearly the same at $n = 100$ and $n = 400$, as an $O(n^{-4})$ remainder requires. That is evidence, not a proof.

[2 October 2026, batch 77P5.] The gap is closed in Part II, for fixed k . Theorem II.3.1 proves the exact factorial-moment bound $\mathbb{E}(X_{n,k})_{\underline{M}} \leq (k-1)^M$ for the present model, the device asked for above, proved directly rather than imported from the path-forest report. Theorem II.4.2 turns it

into an explicit remainder, uniform for $|v| \leq V$; Corollary II.5.1 gives the integer-power expansion without any cancellation of half-powers; and Theorem II.6.1 proves the scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$. Appendix II.A verifies the gamma-tail route of the proof above. Every item listed above as resting on the sketched step is therefore proved for fixed k (Remark II.6.3). Nothing uniform in k is proved; research question 1 of Section 14 is untouched.

7 Exact factorial cumulants

Define factorial cumulants $\kappa_r^{(F)}$ by

$$\log \Phi_{n,k}(v) = \sum_{r \geq 1} \kappa_r^{(F)} \frac{v^r}{r!}. \quad (34)$$

They are particularly natural for a Poisson limit: a Poisson variable of mean λ has $\kappa_1^{(F)} = \lambda$ and all higher factorial cumulants zero.

Theorem 7.1 (First four factorial cumulants). *Let $N = kn$. Whenever the displayed denominators are nonzero,*

$$\kappa_1^{(F)} = k - 1, \quad (35)$$

$$\kappa_2^{(F)} = -\frac{(k-1)^2}{N-1}, \quad (36)$$

$$\kappa_3^{(F)} = \frac{2(k-2)(k-1)^2}{(N-2)(N-1)}, \quad (37)$$

$$\kappa_4^{(F)} = -\frac{6(k-1)^2\{N(k^2-6k+7)+4k-6\}}{(N-3)(N-2)(N-1)^2}. \quad (38)$$

In particular,

$$\boxed{\mathbb{E}X_{n,k} = k - 1, \quad \text{Var } X_{n,k} = (k-1) - \frac{(k-1)^2}{kn-1}}. \quad (39)$$

Proof. Keep terms through v^4 in (20). Since

$$R_k(z) = 1 + a_{k,1}z + a_{k,2}z^2 + a_{k,3}z^3 + a_{k,4}z^4 + O(z^5),$$

ordinary multinomial expansion gives the first four factorial moments as rational functions of n and N . Taking the formal logarithm and using (14) simplifies to (35)–(38). This is finite polynomial algebra; Appendix A records a reproducible symbolic check. For any integer-valued random variable, $\mathbb{E}X = \kappa_1^{(F)}$ and $\text{Var } X = \kappa_2^{(F)} + \kappa_1^{(F)}$, which gives (39). $\square$

All higher factorial cumulants tend to zero. The all-orders theorem gives the sharper scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$ for fixed $r \geq 2$, a mod-Poisson pattern that controls local corrections. [2 October 2026, batch 77P5: this scale does not follow from the existence of the expansion alone; it is proved as Theorem II.6.1 of Part II.]

8 Explicit expansion through third order

Put $\lambda = k - 1$. Expanding the exact cumulants and applying Theorem 6.2 gives the following compact logarithmic form.

Theorem 8.1 (Logarithmic factorial-PGF expansion). *For fixed $k \geq 2$, uniformly for v in compact subsets of $\mathbb{C}$,*

$$\begin{aligned} \log \Phi_{n,k}(v) = & \lambda v - \frac{\lambda^2 v^2}{2kn} \\ & + \frac{\lambda^2 v^2 \{2(k-2)v - 3\}}{6k^2 n^2} \\ & - \frac{\lambda^2 v^2 \{(k^2 - 6k + 7)v^2 - 4(k-2)v + 2\}}{4k^3 n^3} + O_{k,v}(n^{-4}). \end{aligned} \quad (40)$$

Proof. Insert (36)–(38) into (34) and expand their denominators in $1/n$. These terms supply every contribution through n^{-3} because the fifth and higher factorial cumulants begin at $O_k(n^{-4})$, as follows from Theorem 6.2. Collecting powers of v gives (40). [2 October 2026, batch 77P5: the step on the fifth and higher factorial cumulants is Theorem II.6.1 of Part II, and (70) there re-derives the three displayed coefficients from the full operator.] $\square$

Setting $v = -1$ produces the no-adjacency probability.

Corollary 8.2 (No-adjacency transseries). *For fixed $k \geq 2$,*

$$\log p_{n,k} = -(k-1) - \frac{(k-1)^2}{2kn} - \frac{(k-1)^2(2k-1)}{6k^2 n^2} - \frac{(k-1)^4}{4k^3 n^3} + O_k(n^{-4}), \quad (41)$$

$$\begin{aligned} p_{n,k} = e^{-(k-1)} \left[1 - \frac{(k-1)^2}{2kn} + \frac{(k-1)^2(3k^2 - 14k + 7)}{24k^2 n^2} \right. \\ \left. - \frac{(k-1)^4(k^2 - 10k + 17)}{48k^3 n^3} + O_k(n^{-4}) \right]. \end{aligned} \quad (42)$$

The logarithmic form is often numerically superior because exponentiating a truncated log preserves positivity and partially resums products of lower orders.

8.1 A first local correction

The expansion describes more than the mass at zero.

Corollary 8.3 (Fixed local probabilities). *For each fixed $j \geq 0$,*

$$\mathbb{P}(X_{n,k} = j) = e^{-\lambda} \frac{\lambda^j}{j!} \left[1 - \frac{(\lambda - j)^2 - j}{2kn} + O_{k,j}(n^{-2}) \right]. \quad (43)$$

Proof. Exponentiating the first correction in (40) gives

$$\Phi_{n,k}(v) = e^{\lambda v} \left(1 - \frac{\lambda^2 v^2}{2kn} + O(n^{-2}) \right).$$

Write $v = u - 1$ and extract the coefficient of u^j . Dividing the correction by the Poisson mass gives

$$-\frac{\lambda^2}{2kn} \left(1 - \frac{2j}{\lambda} + \frac{j(j-1)}{\lambda^2} \right),$$

which is (43). □

9 Specializations to the OEIS rows

For $k = 2, 3, 4, 5$, (42) becomes

$$p_{n,2} = e^{-1} \left(1 - \frac{1}{4n} - \frac{3}{32n^2} - \frac{1}{384n^3} + O(n^{-4}) \right), \quad (44)$$

$$p_{n,3} = e^{-2} \left(1 - \frac{2}{3n} - \frac{4}{27n^2} + \frac{4}{81n^3} + O(n^{-4}) \right), \quad (45)$$

$$p_{n,4} = e^{-3} \left(1 - \frac{9}{8n} - \frac{3}{128n^2} + \frac{189}{1024n^3} + O(n^{-4}) \right), \quad (46)$$

$$p_{n,5} = e^{-4} \left(1 - \frac{8}{5n} + \frac{8}{25n^2} + \frac{128}{375n^3} + O(n^{-4}) \right). \quad (47)$$

Thus (44) sharpens the known e^{-1} law for A007060 [2], (45) sharpens the e^{-2} law printed on A193624 [3], and (46) supplies explicit corrections for A330266 [1].

9.1 The standard 52-card deck

For $n = 13$ and $k = 4$, exact enumeration gives

$$p_{13,4} = 0.04547628233109430448968591097164118590821040881636\dots \quad (48)$$

The successive approximations are

approximation	numerical value
e^{-3}	0.04978706836786394298
through n^{-1} in (46)	0.04547857206679879407
through n^{-2} in (46)	0.04547166742529067684
through n^{-3} in (46)	0.04547585004466578632
exponential of (41) through n^{-3}	0.04547622509184257975
exact	0.04547628233109430449

The third-order logarithmic approximation has relative error about 1.26×10^{-6} even at the modest value $n = 13$.

The exact mean number of equal-rank adjacencies is 3, while

$$\text{Var } X_{13,4} = 3 - \frac{9}{51} = \frac{48}{17} = 2.8235294117\dots$$

This quantifies the slight underdispersion relative to a $\text{Poisson}(3)$ law.

9.2 Numerical convergence across several multiplicities

The following table lists $e^{k-1}p_{n,k}$, which tends to 1. The final column is the signed error after subtracting all three displayed correction terms in (42).

k	n	$e^{k-1}p_{n,k}$	third-order residual
2	10	0.97406694843814	7.05×10^{-6}
2	50	0.99496248890932	9.74×10^{-9}
3	10	0.93191894503056	1.77×10^{-5}
3	50	0.98660782736509	2.49×10^{-8}
4	10	0.88747575415541	2.56×10^{-5}
4	50	0.97749213853337	3.70×10^{-8}
5	10	0.84356657118181	2.52×10^{-5}
5	50	0.96813076805226	3.74×10^{-8}
6	10	0.80107140086354	1.29×10^{-5}
6	50	0.95869552482316	1.94×10^{-8}

The small residuals are not used as proof; they are independent diagnostics of the symbolic expansion.

10 Asymptotics of the counts, ratios, and inverse functions

10.1 Logarithmic count transseries

Let $N = kn$ and $\lambda = k - 1$. Combining Stirling's expansion with (41) yields

$$\log B_{n,k} = N(\log N - 1) + \frac{1}{2} \log(2\pi N) - \lambda + \frac{A_1}{N} + \frac{A_2}{N^2} + \frac{A_3}{N^3} + O_k(N^{-4}), \quad (49)$$

where

$$A_1 = \frac{1}{12} - \frac{\lambda^2}{2}, \quad A_2 = -\frac{\lambda^2(2k-1)}{6}, \quad A_3 = -\frac{1}{360} - \frac{\lambda^4}{4}. \quad (50)$$

For multiset words,

$$\log S_{n,k} = \log B_{n,k} - n \log(k!). \quad (51)$$

These formulas separate the universal factorial growth from the finite adjacency penalty.

10.2 Successive-term ratios

For fixed k ,

$$\frac{B_{n+1,k}}{B_{n,k}} = \prod_{j=1}^k (kn + j) \left[1 + \frac{(k-1)^2}{2kn^2} + \frac{(k-1)^2(k-2)}{6k^2n^3} + O_k(n^{-4}) \right]. \quad (52)$$

Indeed, the factorial ratio is exact and subtraction of $\log p_{n,k}$ from $\log p_{n+1,k}$ gives the bracket. Such a ratio is useful for checking recurrences attached to individual OEIS rows.

10.3 A Lambert- W inverse transseries

Suppose a large value Y is known and one wants to estimate the index n for which $B_{n,k} \approx Y$. Set

$$L = \log Y + \lambda, \quad N_0 = \frac{L}{W_0(L/e)}. \quad (53)$$

Then $N_0(\log N_0 - 1) = L$. One Newton correction from the remaining terms in (49) gives

$$N = N_0 - \frac{\frac{1}{2} \log(2\pi N_0) + A_1/N_0 + A_2/N_0^2 + A_3/N_0^3}{\log N_0} + O_k(N_0^{-1}), \quad n = \frac{N}{k}. \quad (54)$$

This is an inverse transseries in the same spirit as the inversion examples in the ProveIt repository. Further Newton steps and further terms of Theorem 6.2 extend it mechanically. [Added 1 October 2026, batch 73O2: the inversion examples meant are those of the canonical volume *Transseries and Inversion* [15]. The forward map $\log B_{n,k}$ of (49) is the volume’s gamma carrier (theorem “All-orders expansion of Γ_+^{-1} ”, label `p6:thm:gamma`) perturbed by the power series in $1/N$ of (41), the situation of its theorem “Perturbed inversion” (`p0:thm:perturbed-inversion`); the volume’s chapter “The subfactorial” (`p8:sec:top`) treats the derangement numbers the same way. Normalizations differ: here $L = \log Y + \lambda$ and $N_0 = L/W_0(L/e)$, with $\frac{1}{2} \log(2\pi N_0)$ left in the correction; the volume uses $R = \log y - \frac{1}{2} \log(2\pi)$. Formula (54) is a first-order instance of that apparatus, and no novelty is claimed for it. Deciding the integer index from an approximate inverse, as in the deck example below and in research question 11 of Section 14, is the subject of the volume’s staircase theorem (`p0:thm:staircase`).] [2 October 2026, batch 77P5: retaining A_2, A_3 in a single Newton step does not by itself give the corresponding extra accuracy, since omitted quadratic Newton terms can be larger. Part II proves the sharper error $O_k(1/(N_0 \log N_0))$ along the exact count sequence, (71), which implies the error stated in (54).]

For the exact standard-deck count in (48), with $k = 4$, the leading Lambert estimate gives $n \approx 13.17720934$. The correction (54) gives

$$n \approx 12.99979618,$$

which correctly identifies the integer index 13.

11 Relation to rook polynomials and recent multipartite work

The coefficients $a_{k,m} = m! \binom{k}{m} \binom{k-1}{m}$ are the familiar rook numbers of a rectangular board of dimensions k by $k-1$. This explains why rook-polynomial methods naturally occur in exact deck-shuffle enumeration. Enouen’s report on the standard deck [8] develops such a direct rook-polynomial approach. The reciprocal polynomial $C_k(z) = R_k(-z)$ packages all ranks into the compact power $C_k(z)^n$.

Balanced Smirnov words have also been connected explicitly with Hamiltonian paths and cycles in complete multipartite graphs. Recent work by Mehiri [9] gives bijections and inclusion–exclusion formulas, including nonuniform part sizes. The present asymptotic regime keeps the part size k fixed and lets the number of parts n grow. In this regime, Theorem 5.1 says:

Among all ordered vertex lists of $K_{k,k,\dots,k}$, the fraction that form Hamiltonian paths tends to $e^{-(k-1)}$, and the number of forbidden within-part consecutive steps tends to $\text{Poisson}(k-1)$.

This probabilistic statement complements exact Hamiltonian enumerations and suggests analogous cyclic and nonuniform limit laws.

12 Reproducibility and a formalization route

The accompanying program `code/verify.py` performs the following tasks using exact Python integers for enumeration and high-precision decimal arithmetic for probabilities:

1. constructs $C_k(z) = \sum_m (-1)^m a_{k,m} z^m$;
2. raises it to the n th power by exact polynomial convolution;
3. evaluates (18);
4. checks the initial terms of A007060, A193624, and A330266;
5. verifies the exact 52-card value;
6. writes the numerical convergence table used above.

The optional SymPy script `code/derive_symbolic.py` reconstructs the first four factorial cumulants and checks the displayed n^{-3} expansion. The article does not rely on a fitted numerical constant.

A Lean formalization can be separated into finite and analytic layers. The finite core is especially approachable:

1. define rank words, runs, and the statistic X ;
2. prove the run-compression bijection and finite coefficient identity (8);
3. prove the polynomial identity (11);
4. reduce the integral notation to the factorial moment identity $\int_0^\infty t^m e^{-t} dt = m!$;
5. prove the exact finite sum (18);
6. establish the leading factorial-moment estimate using polynomial degree bounds.

The final method-of-moments step can either use an existing probability convergence library or be replaced by direct convergence of each fixed mass through the probability generating function. The exact count identities can be formalized independently of the asymptotic layer.

13 Proposed OEIS updates

Subject to independent review and attribution, the following concise comments would strengthen the relevant entries.

General fixed- k family. If $B_{n,k}$ counts permutations of kn labeled cards, k in each of n ranks, with no equal adjacent ranks, then for each fixed $k \geq 2$,

$$\frac{B_{n,k}}{(kn)!} = e^{-(k-1)} \left(1 - \frac{(k-1)^2}{2kn} + O_k(n^{-2}) \right).$$

More precisely, the number of equal-rank adjacencies converges to Poisson($k-1$), and (42) gives three correction terms.

A330266. Add (46), together with the exact variance $3 - 9/(4n-1)$ for the adjacency statistic. The integral on the entry is the $k=4$ instance of

$$B_{n,k} = \int_0^\infty \{(-1)^k k! L_k^{(-1)}(t)\}^n e^{-t} dt.$$

A007060 and A193624. Add (44) and (45), respectively. Their multiset partners A114938 and A193638 have the same normalized probabilities because of (3).

A plain-text version is included as `oeis_proposed_updates.txt`; it is explicitly marked as unsubmitted. [Added 1 October 2026, batch 73O2: shipped as `data/oeis_proposed_updates.txt`, unchanged. Nothing has been submitted to the OEIS. Its correction terms rest on the sketched step of Remark 6.3.] [2 October 2026, batch 77P5: that step is now proved for fixed k in Part II (Remark II.6.3); the comments remain unsubmitted.]

14 Further research questions and topics

1. **Uniformly growing multiplicity.** Determine the largest range $k = k(n)$ in which a Poisson approximation remains valid, and locate transitions to compound-Poisson or normal behavior. The fixed- k proof is not uniform when the degree of R_k grows.
2. **Explicit total-variation bounds.** Use Stein–Chen coupling or the gamma representation to obtain a sharp bound of the form $d_{TV}(X_{n,k}, \text{Poisson}(k-1)) \leq C_k/n$, preferably with the optimal constant and a computable second-order signed correction.
3. **Cyclic words and Hamiltonian cycles.** Also forbid equality between the last and first ranks. Find the exact analogue of (40), taking proper account of oriented versus unoriented cycles and rotational symmetries.
4. **Unequal multiplicities.** For ranks of sizes $k_1, \dots, k_n$ and total N , identify natural sparse conditions under which the adjacency count is asymptotically Poisson with mean $\sum_i k_i(k_i-1)/N$. Determine the correction terms in symmetric sums of the k_i .
5. **Larger exclusion distance.** Forbid two equal ranks within distance d rather than only distance one. Cluster expansions should produce a Poisson or compound-Poisson law when k, d are fixed, but overlapping short patterns require new local cumulants.
6. **Moderate and large deviations.** Analyze $\mathbb{P}(X_{n,k} \geq x)$ outside the fixed- x regime. The exact gamma expectation may lead to a saddle-point approximation uniformly for x growing with n .

7. **All-order closed forms.** The operator rule (33) is constructive. Find a canonical Bell-polynomial expression for the coefficient of n^{-r} in $\log \Phi_{n,k}(v)$, and study sign or factorization patterns in k and v . [2 October 2026, batch 77P5: re-scoped. Part II gives the coefficients by the finite formula (66) and proves that the coefficient $q_s(v)$ of N^{-s} has degree at most $s + 1$ in v (Theorem II.6.1); a canonical Bell-polynomial closed form and the sign and factorization patterns remain open.]
8. **Laguerre-zero geometry.** Clarify how the real zeros of $L_k^{(-1)}$ govern recurrences, cancellations, and possible saddle points of the exact integral. The integral involves a sign-changing polynomial, so a naive positive Laplace principle is insufficient.
9. **Differential equations and recurrences.** For each fixed k , derive minimal-order polynomial recurrences for $B_{n,k}$ and compare their asymptotic solutions with (42). Seek bounds on recurrence order as a function of k .
10. **New OEIS rows and arrays.** Create a two-dimensional array for $S_{n,k}$ or $B_{n,k}$, with rows indexed by k , including $k = 5, 6$ and exact kernels P_k . Cross-reference the labeled, multiset, rook-polynomial, and multipartite-graph interpretations.
11. **Rigorous inverse error bounds.** Turn (54) into an explicit bracketing theorem for the least n with $B_{n,k} \geq Y$, including monotonicity thresholds and certified rounding. [2 October 2026, batch 77P5: re-scoped. Part II proves the error $O_k(1/(N_0 \log N_0))$ along the exact count sequence (71), and states explicitly that this certifies no finite integer index; the bracketing theorem asked for here remains open.]
12. **Machine-checked asymptotics.** Formalize the finite identities in Lean, then isolate a reusable theorem that turns fixed factorial-moment convergence plus tightness into Poisson convergence. This would support many related OEIS limit laws.

15 Conclusion

The constant $e^{-(k-1)}$ is not an isolated feature of three OEIS rows. It is the zero mass of a limiting Poisson law for the complete adjacency statistic. The run transform converts the combinatorics into a single Laguerre polynomial; reciprocal coefficients give exact inclusion–exclusion; and a gamma negative-moment representation produces systematic correction terms. The resulting framework simultaneously proves the fixed- k conjecture recorded on A330266, sharpens the known $k = 2, 3, 4$ asymptotics, and opens a tractable family of cyclic, nonuniform, and growing-multiplicity problems.

A Cumulant algebra

For completeness, write

$$R_k(z) = 1 + a_1 z + a_2 z^2 + a_3 z^3 + a_4 z^4 + O(z^5),$$

where

$$\begin{aligned} a_1 &= k(k-1), \\ a_2 &= \frac{k(k-2)(k-1)^2}{2}, \\ a_3 &= \frac{k(k-3)(k-2)^2(k-1)^2}{6}, \\ a_4 &= \frac{k(k-4)(k-3)^2(k-2)^2(k-1)^2}{24}. \end{aligned}$$

The coefficient of v^r in (20) is $[z^r]R_k(z)^n/(N)_{\underline{r}}$. Thus, for example,

$$\begin{aligned} \mathbb{E}(X)_{\underline{2}} &= \frac{2na_2 + n(n-1)a_1^2}{(N)_{\underline{2}}}, \\ \mathbb{E}(X)_{\underline{3}} &= \frac{6na_3 + 6n(n-1)a_2a_1 + n(n-1)(n-2)a_1^3}{(N)_{\underline{3}}}. \end{aligned}$$

The fourth factorial moment is obtained from the five partitions 4, 3 + 1, 2 + 2, 2 + 1 + 1, and 1 + 1 + 1 + 1. Applying the standard moment-to-cumulant relations and simplifying gives (35)–(38). The accompanying SymPy script performs exactly this finite calculation and verifies that the difference from (40) is $O(n^{-4})$.

B Exact initial terms for $k = 5$

The same formula supplies a useful prospective OEIS row. For $n = 0, 1, \dots, 7$,

$$\begin{aligned} B_{n,5} &= 1, \ 0, \ 28800, \ 12420864000, \ 27917203599360000, \\ &\quad 197726965332480000000000, \\ &\quad 3614614673967242167910400000000, \\ &\quad 147444323126376251733903684403200000000. \end{aligned}$$

The kernel is $P_5(t) = t^5 - 20t^4 + 120t^3 - 240t^2 + 120t$, and $B_{n,5} = \int_0^\infty P_5(t)^n e^{-t} dt$. These terms should be independently cross-checked before creating or editing an OEIS entry.

C Provenance and editorial notes

[Added 1 October 2026, batch 73O2.]

Source. The report was built from one manuscript, manuscript 57 of batch 73 of ProveIt’s incoming-reports intake: the archive `ProveIt_Balanced_Smirnov_OEIS.zip` (main file `article.tex`, a 17-page PDF). It arrived in commit `aa43cc555` and was placed in commit `6e193dd4f`, which deleted the archive from `docs/incoming`; it survives in `aa43cc555`. The manuscript pins no repository revision and cites no repository file. It has no author or preparer line: its author field is the subtitle “A research note on OEIS A007060, A193624, A330266, and related sequences”, and the delivered PDF’s author metadata is empty.

What the write changed.

- Every label gained the prefix `bsw:`; none was otherwise renamed or removed. The manuscript had 66 labels.
- Citation commands were added: the manuscript listed twelve bibliography entries but cited none of them. They now stand at the first mentions of the OEIS entries, the Smirnov-word generating function, the Laguerre expansion, and the works of Eriksson–Martin, Enouen and Mehiri.
- The authors and dates of the two OEIS contributions are named: S. Kirgizov’s fixed- k conjecture (29 September 2023) and D. Radcliffe’s deduction of the case $k = 4$ (9 September 2025).
- Added: the editorial note after the scope paragraph, Remark 6.3 on the sketched analytic step, a note in the inverse section citing the transseries volume [15], a note on the shipped file of proposed OEIS comments, one bibliography entry, the label `bsw:sec:questions`, and this section.
- Nothing was removed. Every non-claim of the manuscript is kept: no complete literature search or priority claim, the numerical residuals “not used as proof”, the cross-check required for the $k = 5$ terms, and the unsubmitted status of the proposed OEIS comments.

Relation to the companion report. The path-forest report `a189281-path-forest-expansions` of this collection (batch 73, manuscript 60) follows the same chain of method (exact marked-subset generating function, Poisson limit, all-orders $1/n$ expansion of a factorial probability generating function, Lambert- W inversion) for permutations avoiding a fixed positional and value offset, with limit $\text{Poisson}(\theta)$, $\theta \in \{1, 2\}$. Neither theorem specializes to the other. In the present model the n ranks give n cliques K_k as the “value” relation, whose number grows with n , so the hypotheses of the path-forest report (a fixed number of long paths) do not hold. Its uniform factorial-moment bound is the device named in Remark 6.3.

Checks by the intake. The intake reran both shipped programs on a copy (the CSV is byte-identical; the deck output equal apart from line endings; the symbolic script confirms $\kappa_1^{(F)}, \dots, \kappa_4^{(F)}$ and the n^{-3} expansion), compared the terms of A007060, A193624 and A330266 with an independent dynamic program over rank words, confirmed the $k = 5$ terms (for $n \leq 4$ by that program, for $n \leq 7$ by the formula) and the 52-card value, checked the variances $4/5, 6/5, 6/7$ at $(n, k) = (3, 2), (2, 3), (4, 2)$ by full enumeration, and rederived by hand the local correction, the ratio bracket and A_1, A_2, A_3 . These checks do not replace an independent review of the proofs.

Part II. [Added 2 October 2026, batch 77P5.] Part II was added from a second manuscript, manuscript 11 of batch 77, which pins this report at commit `4b874cea0`; its provenance, the choices of the merge and the intake’s checks are in Section II.9. Besides Part II, the batch-77 write added the dated notes of 2 October 2026 in this Part and two bibliography entries (Gessel, Taylor), and changed no earlier text.

Part II

A rigorous uniform remainder for balanced Smirnov asymptotics

Editorial note (2 October 2026, batch 77P5). Part II is manuscript 11 of batch 77 of ProveIt’s incoming-reports intake, *A rigorous uniform remainder for balanced Smirnov asymptotics*, dated 2 October 2026, whose author line reads “Mathematical repair and reproducibility report”. It arrived in the archive `balanced-smirnov-repair-reproducibility.zip`; AI-assisted, unrefereed, not formalized. It audits this report as it stood at commit `4b874cea0` and proves the analytic step that Remark 6.3 records as a sketch. Sections 1–15 and Appendices A–C above form Part I, unchanged except for notes dated 2 October 2026. The write prefixed every label of the manuscript with `bsw:rp:`, replaced its citations of “Theorem 6.2”, “Remark 6.3” and “Theorem 4.1” of the pinned source by cross-references, renamed the symbols listed in Section II.1, printed once the formulas that Part I already displays (Section II.7), and collected the source pin and the provenance in Section II.9.

Summary (the manuscript’s abstract). For a uniform shuffle of n ranks of k cards, with fixed $k \geq 2$, let X count equal-rank adjacencies and put $N = kn$. We prove the exact bound $\mathbb{E}(X)_{\underline{M}} \leq (k-1)^M$ and use it to justify the full factorial-probability-generating-function expansion uniformly on compact subsets of the complex plane. The remainder has an explicit Poisson-weighted bound. A separate formal-operator argument proves the degree cancellations needed for all-orders factorial-cumulant scaling. This closes the analytic gap in Theorem 6.2 of Part I. We verify its displayed third-order formulas, give the local, count and ratio consequences, and state the inverse approximation with its necessary scope. The existing combinatorial formulas and displayed coefficients are credited to their sources; no claim of historical novelty is made.

II.1 Notation of Part II

[Added at the write, 2 October 2026.] Part II keeps Part I’s notation wherever the two agree: $N = kn$, $\lambda = k-1$, $a_{k,m}$ and $R_k(z)$ of (14)–(15), $(x)_{\underline{M}}$, $p_{n,k}$, $B_{n,k}$, $S_{n,k}$, $H_r(M)$ of (28), $D = v d/dv$, the factorial cumulants $\kappa_r^{(F)}$ of (34), A_1, A_2, A_3 of (50), and $L = \log Y + \lambda$, N_0 of (53). The manuscript wrote Φ_n for Part I’s $\Phi_{n,k}$ of (19) and e for $\mathbf{e}$; Part II uses Part I’s forms. The other symbols of the manuscript that collide with Part I, or with each other, were renamed as follows.

Part II	manuscript	meaning, and the reading to avoid
$c_{n,M}$	$c_{n,M}$	$\mathbb{E} \binom{X}{M} = r_{n,M}/(N)_{\underline{M}}$, with $r_{n,M} = [z^M]R_k(z)^n$ as in the proof of Theorem 6.2; not the carrier constant c_k of Section II.7.3
d	d	$n(k-1)$, the largest possible value of X ; not the d of d/dv
$\sigma_j(M)$	$p_j(M)$	the power sum $\sum_{i \leq M} i^j$; not the probability $p_{n,k}$
$F_N(v)$	$F_N(v)$	$R_k(v/N)^n$, the bracket in (33); not the gamma density $f_N(t)$ of (25)
$\mathcal{P}_s(v)$	$P_s(v)$	the coefficient polynomials of $e^{-\lambda v} \Phi_{n,k}(v)$ in powers of N^{-1} ; <i>not</i> the Laguerre kernels $P_k(t)$ and $P_{k,u}(t)$ of (9)–(13): $P_2(v) \neq P_2(t) = t^2 - 2t$
J	L	the truncation order; L stays $\log Y + \lambda$ as in (53)
$\mathcal{C}_{k,V,J}$	$C_{k,V,L}$	the remainder constant of Theorem II.4.2; not the reciprocal polynomial $C_k(z)$ of (16)
$\mathcal{T}_j(a)$	$\mathcal{T}_j(a)$	the j th Touchard polynomial; not the gamma variable T_N
$\mathcal{H}_w$	T_w	the formal operator $\sum_r H_r(D)w^r$; at $w = 1/N$ it is an operator, <i>not</i> the gamma variable T_N of (25)
$\mathcal{A}_w(D)$	$A_w(D)$	the generator $\sum_{j \geq 1} w^j \sigma_j(D)/j$; not the Stirling constants A_1, A_2, A_3
$\Sigma(t, w, v), \Sigma_s(t, v)$	S, S_s	the formal logarithm along the operator flow; not the multiset count $S_{n,k}$
$\mathcal{B}_a$	B_a	$e^{-\Sigma} D^a e^{\Sigma}$; not the count $B_{n,k}$
γ_j	c_j	$(b_{j+1}/k)v^{j+1}$, an abbreviation in Section II.7
$\psi(x), g(x)$	$f(x), h(x)$	the inverse carrier $x(\log x - 1)$ and its correction; renamed because f_s and h_s denote other objects
M_N	J	the cut $\lfloor N^{1/3} \rfloor$ of Appendix II.A

Normalization: the manuscript expands in powers of $N^{-1} = (kn)^{-1}$, Part I in powers of n^{-1} ; a coefficient of N^{-s} is k^{-s} times the corresponding coefficient of n^{-s} . Two readings are excluded explicitly below: Theorem II.3.1 is factorial-moment domination, not stochastic domination by a Poisson variable; and $\kappa_r^{(F)}$ are *factorial* cumulants, not ordinary cumulants (Remark II.6.2).

II.2 Scope and the gap being repaired

Part I considers *fixed multiplicity*: k is fixed while the number n of ranks tends to infinity. Its Theorem 6.2 gives a formal coefficientwise operator expansion, but the transition to compact-complex-uniform asymptotics sketches three steps: weighted gamma-tail control, suppression of large coefficient degrees, and cancellation of odd half-powers. Its editorial Remark 6.3 explicitly identifies this gap. The exact generating functions, exact first four factorial cumulants, Poisson limit and leading no-adjacency limit do not require that step.

The proof below supplies a direct quantitative remainder, avoiding a saddle expansion. It also supplies a separate all-orders logarithmic degree argument: existence of a uniform expansion alone would not prove the asserted factorial-cumulant scaling. Appendix II.A additionally verifies the weighted gamma-tail assertion. No long-path hypothesis from the neighboring path-forest report `a189281-path-forest-expansions` is used; its theorem cannot simply be imported into this growing collection of rank cliques.

All asymptotic statements below fix k . Constants may depend on k , the compact set and the truncation order. Nothing here asserts uniformity for growing k , or a uniform relative local

approximation for growing adjacency count.

II.3 Exact model and factorial-moment domination

There are $n \geq 1$ ranks, each with $k \geq 2$ distinct cards. Choose a permutation of the $N = kn$ cards uniformly. Let $X = X_{n,k}$ be the number of positions $j < N$ at which cards $j, j+1$ have the same rank. Equivalently choose a multiset word uniformly with k copies of each of n letters. Write

$$\lambda = k - 1, \quad d = n(k - 1), \quad \Phi_{n,k}(v) = \mathbb{E}(1 + v)^X, \quad p_{n,k} = \mathbb{P}(X = 0).$$

The labeled no-adjacency count is $B_{n,k} = N!p_{n,k}$; the multiset count is $S_{n,k} = B_{n,k}/(k!)^n$. We have $0 \leq X \leq d < N$. For $k = 2, 3, 4$, respectively, the labeled counts are OEIS A007060, A193624 and A330266 [2, 3, 1]; the corresponding multiset-word counts are A114938, A193638 and A321633 [4, 5, 6]. The probability is common to both models, but the counts are different.

With $(a)_0 = 1$, $a_{k,m}$ and R_k as in (14)–(15) (the manuscript writes $a_{k,m} = \binom{k-1}{m}(k)_m$, the same number), put

$$r_{n,M} = [z^M]R_k(z)^n, \quad c_{n,M} = \mathbb{E}\binom{X}{M} = \frac{r_{n,M}}{(N)_M} \quad (0 \leq M \leq d). \quad (55)$$

Set $r_{n,M} = c_{n,M} = 0$ for $M > d$; do not evaluate the quotient outside its stated support. The exact identity

$$\Phi_{n,k}(v) = \sum_{M=0}^d \frac{[z^M]R_k(z)^n}{(N)_M} v^M$$

is (20) of Theorem 4.1, which Part I obtains by weighted run transforms.

Second proof of (20) (manuscript 11). For completeness, there is a direct contraction proof. Canonically number the occurrences of each letter from left to right. The $k - 1$ possible bonds for a letter join consecutive occurrences; a bond is present when those occurrences are adjacent. Select m_i bonds for letter i and contract them. The contracted words have $k - m_i$ copies of letter i , and expansion according to the selected components is inverse to contraction. If $M = \sum_i m_i$, the number of such words is $(N - M)! / \prod_i (k - m_i)!$. Double counting words with M selected present bonds and dividing by $N!/(k!)^n$ gives (55). $\square$

The Laguerre and rook-polynomial machinery is older: Gessel's linear-permutation construction [10] (its Section 6 treats linear permutations), Taylor's Theorem 2.1 and run generating function [11] (his Theorem 2.1 credits Gessel's exact Carlitz-word formula; his Lemma 2.2 gives the classical generating function), and Enouen's standard-deck application [8] (its Sections 3–4 give the rook-polynomial solution and the exact standard-deck count) give relevant precedents. Eriksson and Martin [7] enumerate the same Carlitz multipermutation family. This Part repairs a specified proof, rather than claiming priority for those formulas.

Theorem II.3.1 (Exact Poisson factorial domination). *For all $n \geq 1$ and integers $M \geq 0$,*

$$0 \leq c_{n,M} \leq \frac{\lambda^M}{M!}, \quad \text{equivalently} \quad \mathbb{E}(X)_M \leq \lambda^M. \quad (56)$$

Consequently $|\Phi_{n,k}(v)| \leq e^{\lambda|v|}$ for every complex v .

Proof. Since $(k)_{\underline{m}} \leq k^m$, nonnegative coefficients give

$$R_k(z) \preceq (1 + kz)^{k-1}, \quad r_{n,M} \leq k^M \binom{d}{M}.$$

Here $\preceq$ means coefficientwise inequality. For $0 \leq M \leq d$,

$$c_{n,M} \leq \frac{k^M}{M!} \prod_{j=0}^{M-1} \frac{d-j}{N-j} \leq \frac{k^M}{M!} \left(\frac{d}{N}\right)^M = \frac{\lambda^M}{M!},$$

since $(d-j)/(N-j) \leq d/N$. Empty products include $M = 0$, and the zero convention includes $M > d$. Sum the absolute coefficients to obtain the last assertion. $\square$

Remark II.3.2. This is factorial-moment domination; it is not a claim of stochastic domination by a Poisson random variable. It includes $n = 1$, when $X = k - 1$ deterministically. [Added at the write: it is the analogue, for the present model, of the uniform factorial-moment bound of the path-forest report that Remark 6.3 names, proved here directly.]

II.4 The explicit uniform operator remainder

For a finite list of nonnegative variables let h_s denote the complete homogeneous symmetric polynomial. Set $h_0 = 1$; for the empty list set $h_s = 0$ when $s > 0$. For integers $M, r \geq 0$ define

$$H_r(M) = h_r(0, 1, \dots, M-1), \quad Q_M(x) = \prod_{j=0}^{M-1} (1 - jx)^{-1},$$

in agreement with (28). The list for $M = 0$ is empty. Both $M = 0$ and $M = 1$ give $Q_M = 1$ and $H_r(M) = 0$ for $r \geq 1$.

Lemma II.4.1 (Complete-homogeneous remainder). *For nonnegative integers a, b , $h_{a+b} \leq h_a h_b$ on nonnegative variables. Consequently, for an integer $J \geq 0$ and $0 \leq x < 1/(M-1)$ when $M \geq 2$,*

$$0 \leq Q_M(x) - \sum_{r=0}^J H_r(M) x^r \leq x^{J+1} H_{J+1}(M) Q_M(x). \quad (57)$$

The inequality holds without restriction on $x \geq 0$ for $M = 0, 1$. For every integer $s \geq 1$,

$$H_s(M) \leq \left(\frac{M(M-1)}{2} \right)^s \leq M^{2s}. \quad (58)$$

Proof. A term of h_{a+b} is indexed by a nondecreasing string of $a+b$ indices. Splitting after a entries injects its terms, with unchanged weights, into the pairs of strings counted by $h_a h_b$. Thus $H_{J+1+t}(M) \leq H_{J+1}(M) H_t(M)$. The convergent expansion $Q_M(x) = \sum_{r \geq 0} H_r(M) x^r$ now gives (57) by summing over $t \geq 0$. Finally, the multinomial expansion of the sum of the variables raised to s counts each weak monomial with coefficient at least one; this proves (58). $\square$

Put $F_N(v) = R_k(v/N)^n$. The symbol $H_r(D)$ denotes a polynomial in the Euler differential operator $D = v d/dv$. Indeed, if

$$\sigma_j(M) = \sum_{i=0}^{M-1} i^j,$$

then

$$\sum_{r \geq 0} H_r(M) x^r = \exp \left(\sum_{j \geq 1} \frac{x^j}{j} \sigma_j(M) \right). \quad (59)$$

Faulhaber's formula gives $\deg \sigma_j = j + 1$; coefficient extraction shows that $H_r(M)$ is a polynomial in M of degree at most $2r$, with the stated values also at $M = 0$.

Theorem II.4.2 (Quantitative compact-complex remainder). *For every integer $J \geq 0$ and real $V \geq 0$, for all $n \geq 1$,*

$$\sup_{|v| \leq V} \left| \Phi_{n,k}(v) - \sum_{r=0}^J N^{-r} H_r(D) F_N(v) \right| \leq \frac{\mathcal{C}_{k,V,J}}{N^{J+1}}, \quad (60)$$

where, with $a = \lambda V$,

$$\mathcal{C}_{k,V,J} = \sum_{M=0}^{\infty} H_{J+1}(M) \frac{a^M}{M!} \leq e^a \mathcal{T}_{2J+2}(a) < \infty. \quad (61)$$

Here $\mathcal{T}_j(a) = e^{-a} \sum_{M \geq 0} M^j a^M / M!$ is the j th Touchard polynomial. For $V = 0$ the constant is zero.

Proof. The exact finite identity (20) reads

$$\Phi_{n,k}(v) = \sum_{M=0}^d r_{n,M}(v/N)^M Q_M(1/N).$$

Every M in this sum satisfies $M \leq d < N$, so Lemma II.4.1 applies. Since $D(v^M) = Mv^M$, subtract the finite operator sum, take absolute values, and use nonnegativity to obtain

$$\begin{aligned} \text{error} &\leq N^{-J-1} \sum_{M=0}^d H_{J+1}(M) r_{n,M}(V/N)^M Q_M(1/N) \\ &= N^{-J-1} \sum_{M=0}^d H_{J+1}(M) c_{n,M} V^M \\ &\leq N^{-J-1} \mathcal{C}_{k,V,J}. \end{aligned}$$

The last step uses Theorem II.3.1. The bound (58) proves finiteness and the Touchard upper bound. In particular there is no need to control $Q_M(1/N)$ uniformly in M : its entire value converts the majorant back into an exact factorial-moment coefficient. $\square$

II.5 Integer powers and the normalized logarithm

Choose the analytic germ

$$\log R_k(z) = \sum_{j \geq 1} b_j z^j, \quad \log R_k(0) = 0, \quad b_1 = k\lambda.$$

For w near zero set

$$F(w, v) = \exp \left(\frac{\log R_k(wv)}{kw} \right), \quad F(0, v) = e^{\lambda v}. \quad (62)$$

On each fixed v -disk the singularity at $w = 0$ is removable and the function is jointly analytic for sufficiently small $|w|$. At $w = 1/N$ it equals $F_N(v)$ exactly, since $N/k = n$ is integral. Its expansion is

$$F(w, v) = e^{\lambda v} \sum_{s \geq 0} f_s(v) w^s, \quad f_0 = 1, \quad (63)$$

$$f_s(v) = \sum_{\substack{m_1, \dots, m_s \geq 0 \\ \sum j m_j = s}} \prod_{j=1}^s \frac{((b_{j+1}/k)v^{j+1})^{m_j}}{m_j!}. \quad (64)$$

Thus f_s is a polynomial of degree at most $2s$. Analyticity on a slightly larger v -disk and Cauchy's estimates justify differentiating each fixed-order Taylor remainder finitely many times, uniformly on $|v| \leq V$.

Corollary II.5.1 (All-orders expansion). *For each fixed integer $J \geq 0$,*

$$\Phi_{n,k}(v) = e^{\lambda v} \sum_{s=0}^J \mathcal{P}_s(v) N^{-s} + O_{k,V,J}(N^{-J-1}), \quad |v| \leq V, \quad (65)$$

where $\mathcal{P}_0 = 1$ and

$$\mathcal{P}_s(v) = e^{-\lambda v} \sum_{r=0}^s H_r(D) (e^{\lambda v} f_{s-r}(v)). \quad (66)$$

Each $\mathcal{P}_s$ is a polynomial in v , with rational-in- k coefficients.

Proof. In Theorem II.4.2, expand the r th differentiated bracket through w^{J-r} . Its multiplied remainder is $O(w^{J+1})$, uniformly on the disk. Summing finitely many such remainders gives (65). All powers are integer powers from analytic Taylor series; no half-power cancellation is assumed. $\square$

On a fixed disk, $e^{-\lambda v} \Phi_{n,k}(v) = 1 + O(N^{-1})$. Hence for sufficiently large n it lies within distance $1/2$ of 1, on a neighborhood of that disk. Define

$$\log \Phi_{n,k}(v) := \lambda v + \log(e^{-\lambda v} \Phi_{n,k}(v)), \quad (67)$$

where the second logarithm is the analytic branch near 1. This branch has value zero at $v = 0$. It has a uniform integer-power expansion. An unspecified principal logarithm on a large complex disk would not give the same statement.

II.6 The all-orders factorial-cumulant degree argument

The next argument is separate from existence of the uniform expansion. It proves the degree cancellations needed to control all factorial cumulants.

Work formally in w ; all operator expressions have finitely many contributions to any fixed coefficient. By (59),

$$\mathcal{H}_w := \sum_{r \geq 0} H_r(D) w^r = \exp \mathcal{A}_w(D), \quad \mathcal{A}_w(D) = \sum_{j \geq 1} \frac{w^j}{j} \sigma_j(D). \quad (68)$$

This may be checked on each monomial v^M using the scalar identity for $Q_M(w)$. No convergence of the infinite operator series at nonzero w is asserted.

Introduce a formal flow parameter t and write

$$e^{t\mathcal{A}_w(D)}F(w, v) = e^{\Sigma(t, w, v)}, \quad \Sigma(t, w, v) = \lambda v + \sum_{s \geq 1} \Sigma_s(t, v)w^s.$$

After removal of $e^{\lambda v}$, this is a formal w -series with polynomial coefficients in v, t and constant term 1; its formal logarithm is uniquely defined. Initially,

$$\Sigma_s(0, v) = \frac{b_{s+1}}{k}v^{s+1}.$$

Theorem II.6.1 (Logarithmic degree and factorial cumulants). *The expansion of the normalized logarithm (67) has the form*

$$\log \Phi_{n,k}(v) = \lambda v + \sum_{s=1}^J q_s(v)N^{-s} + O_{k,V,J}(N^{-J-1}), \quad \deg_v q_s \leq s+1. \quad (69)$$

Consequently, for each fixed $r \geq 2$,

$$\kappa_r^{(F)} := \left. \frac{d^r}{dv^r} \log \Phi_{n,k}(v) \right|_{v=0} = O_{k,r}(N^{1-r}) = O_{k,r}(n^{1-r}).$$

Also $\kappa_1^{(F)} = \mathbb{E}X = \lambda$ exactly.

Proof. The formal flow satisfies

$$\partial_t \Sigma = e^{-\Sigma} \mathcal{A}_w(D) e^{\Sigma},$$

where the conjugated operator on the right acts on 1. Put $\mathcal{B}_a = e^{-\Sigma} D^a e^{\Sigma}$. Then $\mathcal{B}_0 = 1$ and

$$\mathcal{B}_{a+1} = D\mathcal{B}_a + (D\Sigma)\mathcal{B}_a.$$

Thus each monomial of $\mathcal{B}_a$ is a product of at most a factors $D^\ell \Sigma$, $\ell \geq 1$, apart from the constant for $a = 0$.

We induct on s to prove $\deg_v \Sigma_s(t, v) \leq s+1$. To extract w^s from the j th summand of the generator, $1 \leq j \leq s$, the orders s_i of these factors sum to $s-j$. Each is strictly less than s . Because σ_j has degree at most $j+1$, there are at most $b \leq j+1$ factors. The Euler derivative does not increase polynomial degree. The induction hypotheses, including $\Sigma_0 = \lambda v$, give total degree at most

$$\sum_{i=1}^b (s_i + 1) = s - j + b \leq s + 1.$$

The zero-factor case causes no problem. Therefore $\partial_t \Sigma_s$ has this degree bound, and the initial value does too. Integration in t proves the induction. At $t = 1$, the coefficients are those of $\mathcal{H}_w F$, which Corollary II.5.1 has already identified as the analytic asymptotic expansion of $\Phi_{n,k}$. Finite logarithmic-series algebra identifies $q_s = \Sigma_s(1, v)$.

For a fixed $r \geq 2$, take $J = r - 2$ in (69) on a disk about zero. All retained polynomials have degree less than r . Cauchy's coefficient formula applied to the uniform remainder gives the claimed bound. The exact mean follows from (55) with $M = 1$. $\square$

Remark II.6.2. These are *factorial* cumulants, using derivatives in v of $\log \mathbb{E}(1 + v)^X$. Ordinary cumulants use $\log \mathbb{E}e^{tX}$ and converge to the Poisson value λ ; they do not have the displayed decay.

Remark II.6.3 (What this settles in Part I). [Added at the write.] Theorem II.4.2 and Corollary II.5.1 prove Theorem 6.2: (60) is the operator rule (33) with an explicit remainder, uniform for $|v| \leq V$. Theorem II.6.1 proves the scale $\kappa_r^{(F)} = O_{k,r}(n^{1-r})$ stated after Theorem 7.1, and with it the step “the fifth and higher factorial cumulants begin at $O_k(n^{-4})$ ” in the proof of Theorem 8.1 ($r = 5$ gives $O_k(n^{-4})$). Hence every item that Remark 6.3 lists as resting on the sketched step is now proved, for fixed k ; nothing uniform in k is claimed.

II.7 Verified formulas and consequences

The coefficients in this section are those displayed in Part I; the contribution here is the proved remainder and independent algebraic verification. The reproducibility scripts compute them from the full operator, rather than assuming that four cumulants exhaust third order. [Write: formulas that Part I already displays are cited, not printed a second time; the manuscript reprinted them in the variable N , and they agree term by term.]

The first kernel-log coefficients are

$$\begin{aligned} b_1 &= k\lambda, & b_2 &= -k\lambda^2, \\ b_3 &= \frac{k\lambda^2(5k-6)}{3}, & b_4 &= -\frac{k\lambda^2(7k^2-18k+12)}{2}. \end{aligned}$$

If $\gamma_j = (b_{j+1}/k)v^{j+1}$, then

$$f_1 = \gamma_1, \quad f_2 = \gamma_2 + \frac{1}{2}\gamma_1^2, \quad f_3 = \gamma_3 + \gamma_1\gamma_2 + \frac{1}{6}\gamma_1^3.$$

Use $e^{-\lambda v}D(e^{\lambda v}p) = vp' + \lambda vp$ in (66); logarithmic-series algebra gives

$$q_1 = -\frac{\lambda^2 v^2}{2}, \quad q_2 = \frac{\lambda^2 v^2 \{2(k-2)v-3\}}{6}, \quad q_3 = -\frac{\lambda^2 v^2 \{(k^2-6k+7)v^2-4(k-2)v+2\}}{4}, \quad (70)$$

so that (69) with $J = 3$ and $N = kn$ is exactly (40) of Theorem 8.1, with the branch (67) and uniformity on $|v| \leq V$. This verifies the cancellations of possible v^5, v^6 terms directly.

II.7.1 No-adjacency probability and local masses

Setting $v = -1$ gives (41) and (42) of Corollary 8.2; the verified coefficients of n^{-1}, n^{-2}, n^{-3} in $e^{k-1}p_{n,k}$ for $k = 2, 3, 4, 5$ are those of (44)–(47), which the manuscript tabulates again. For each fixed integer $j \geq 0$, Cauchy’s formula on a fixed circle in u , with $v = u - 1$, gives (43) of Corollary 8.3.

Second derivation of the local correction (manuscript 11). The first correction is $-\lambda^2(u-1)^2/(2N)$ times the Poisson PGF. Dividing its u^j coefficient by the Poisson mass gives $-\lambda^2 - 2\lambda j + j(j-1)/(2N)$, including $j = 0, 1$. $\square$

II.7.2 Counts and successive ratios

Stirling’s formula and (41) yield (49), with the constants A_1, A_2, A_3 of (50). The multiset count satisfies $\log S_{n,k} = \log B_{n,k} - n \log(k!)$ exactly (51). In particular its leading growth is $N(\log N - 1 - \log(k!)/k)$, rather than the labeled carrier in (49). For fixed k the ratio expansion (52) holds.

Second proof of (52) (manuscript 11). To see this without differentiating a discrete remainder, if $\log p_{n,k} = -\lambda + a/n + b/n^2 + c/n^3 + O(n^{-4})$, subtraction at consecutive integers gives $-a/n^2 + (a - 2b)/n^3 + O(n^{-4})$. Exponentiation does not change those two terms, and the factorial ratio is exact. $\square$

II.7.3 Carefully scoped Lambert inverse

This subsection inverts the labeled count $B_{n,k}$, not the multiset count $S_{n,k}$. For $Y = B_{n,k}$ along the exact count sequence, define L and N_0 as in (53):

$$L = \log Y + \lambda, \quad N_0 = \frac{L}{W_0(L/e)}.$$

Thus $N_0(\log N_0 - 1) = L$. With

$$g(x) = \frac{1}{2} \log(2\pi x) + A_1/x + A_2/x^2 + A_3/x^3,$$

the following holds as $n \rightarrow \infty$:

$$N = N_0 - \frac{g(N_0)}{\log N_0} + O_k\left(\frac{1}{N_0 \log N_0}\right), \quad n = N/k. \quad (71)$$

In particular it implies the coarser $O_k(N_0^{-1})$ error of (54). Put $\psi(x) = x(\log x - 1)$. Equation (49) says $\psi(N) + g(N) = \psi(N_0) + O(N^{-4})$. First $N \sim N_0$; then $\psi' = \log x$ and $g(N) = O(\log N)$ give $N - N_0 = O(1)$. Taylor expansion of these explicit smooth functions, using $\psi''(x) = 1/x$ and $g'(x) = O(1/x)$, gives

$$(N - N_0) \log N_0 + g(N_0) = O(1/N_0),$$

which proves (71) without differentiating the discrete error.

For arbitrary Y off the sequence, a continuous interpolation or an integer-threshold convention must be specified. Asymptotic notation alone does not certify recovery of a particular finite integer index. Retaining A_2, A_3 in a single Newton correction does not by itself yield the corresponding extra inversion accuracy: omitted quadratic Newton terms can be larger. For a multiset-count inverse, the leading carrier instead uses $c_k = 1 + \log(k!)/k$ and $N_0 = L/W_0(Le^{-c_k})$, with Newton denominator $\log N_0 - \log(k!)/k$. The same proof applies after making those replacements. The inversion apparatus itself is not claimed to be new. [Added at the write: as the note in Part I's inverse section says, the forward map is the gamma carrier of the transseries volume [15] (`p6:thm:gamma`) perturbed by a power series in $1/N$, the setting of its theorem “Perturbed inversion” (`p0:thm:perturbed-inversion`); the carrier equation $\psi(N_0) = L$ is solved by W_0 directly rather than through the dominant-block form $aX + b \log X = L$ of `p0:thm:lambert-core`. The interpolation and integer-threshold conventions asked for above are the subject of its staircase theorem (`p0:thm:staircase`). No novelty is claimed for (71).]

II.8 Reproducibility and review

[Rewritten at the write to name the shipped files.] The delivered archive contained the editable $\text{\LaTeX}$ source and a 10-page PDF of this Part, two independent Python checks, their exact outputs, a mathematical audit, provenance, replay instructions and a clean replay script. It does not redistribute the source manuscript (Part I) or third-party papers; pinned links and hashes are provided instead. This report ships the following delivered files.

shipped file	content
<code>code/02-repair-check_operator.py</code>	the operator check
<code>code/02-repair-check_independent.py</code>	the independent check
<code>code/02-repair-replay.sh</code>	the replay script
<code>data/02-repair-operator.txt</code>	recorded output of the operator check
<code>data/02-repair-independent.txt</code>	recorded output of the independent check
<code>data/02-repair-clean-replay.txt</code>	record of the delivery's clean replay
<code>data/02-repair-environment.txt</code>	the delivery's software environment
<code>data/02-repair-requirements.txt</code>	the SymPy pin
<code>data/02-repair-sources.json</code>	source pin, checksums, literature links
<code>02-repair-mathematical-audit.md</code>	the mathematical audit
<code>02-repair-integrated-review.md</code>	the integrated review
<code>02-repair-visual-qa.md</code>	visual check of the delivered PDF

The manuscript source and PDF (merged here as Part II), the PDF build script, its build log and the checksum ledger are not shipped; the README says how to rerun the checks.

The operator check constructs b_j , f_s , the complete operators $H_r(D)$ and the logarithm through order N^{-3} , and asserts the displayed identities. The independent check enumerates adjacency histograms by exact multiset-word dynamic programming for $k = 2, \dots, 5$, $n = 1, \dots, 5$; verifies every factorial moment through two indices past support; checks the remainder inequalities with rational arithmetic; and independently derives the symbolic coefficients. All assertions passed with SymPy 1.14.0. Finite checks support reproducibility but are not substitutes for the proofs above.

The accompanying mathematical audit found no substantive defect in the uniform remainder, formal degree argument, dependent formulas or optional gamma-tail proof. This is an AI-assisted, independently cross-checked mathematical report, not a peer-reviewed publication or a formal proof-assistant development. The targeted literature check is not exhaustive and establishes no historical novelty.

II.9 Provenance of Part II

[Added 2 October 2026, batch 77P5; the first paragraph is the manuscript's "Source pin" appendix.]

Source pin. The audited ProveIt commit is

4b874cea0012c51a6841ad9c58f6e20fa57c71da.

The report directory is that of this report. The provenance file `data/02-repair-sources.json` gives its complete repository path, full pinned URLs and SHA-256 checksums for the inspected `article.tex` and editorial `README.md`. The full third-party source files are not bundled. [Write: those checksums identify the two files as they stood at 4b874cea0, which is also their state after the batch-73O2 write 728773045; this write changed both files, so the checksums no longer match the current ones.]

Intake record. Manuscript 11 of batch 77 (cluster P5), from the archive named in the editorial note at the start of this Part; arrival commit 096ee7b87; placement commit 4f11bc9c0, which staged twelve delivered files with the prefix `02-repair-` and removed the archive from `docs/incoming` (it survives in 096ee7b87). The manuscript's own pin is 4b874cea0. Its bibliography item for the source report is this report itself and is replaced by internal references; its items for Eriksson–Martin and Enouen are Part I's [7, 8]; Gessel [10] and Taylor [11] were added to the bibliography.

Where the merge had to choose.

- Formulas already displayed in Part I (the third-order logarithmic expansion, $\log p_{n,k}$, $p_{n,k}$, the $k = 2, \dots, 5$ coefficients, the local masses, $\log B_{n,k}$ with A_1, A_2, A_3 , the ratio) are cited, not reprinted; the manuscript's derivations of them are kept, and its contraction proof of (20), its ratio argument and its local computation appear as marked second proofs. The new polynomials q_1, q_2, q_3 are printed in (70).
- Symbols were renamed as listed in Section II.1; no formula changed its content.
- The manuscript's Section 7 described its archive; it is rewritten to name the shipped files.
- Nothing of the manuscript's mathematics, scope statements or disclaimers was removed.

Checks by the intake. The intake reran both checks on a copy (SymPy 1.14.0 under Python 3.13.5 and 3.14.4 on Windows); their outputs equal the recorded ones apart from line endings. Independently of the delivered scripts it confirmed $c_{n,M} \leq \lambda^M/M!$ in exact arithmetic for $k = 2, \dots, 7$, $n \leq 40$; the inequality of Theorem II.4.2 at sixteen points of $|v| = 2$ for $k = 2, 4$, $n = 5, 20$, $J = 0, 1, 2$; that $\deg_v q_s = s + 1$ exactly for $s = 1, \dots, 4$ and $k = 3, 4$ (so the degree bound of Theorem II.6.1 is attained); that q_3 equals the n^{-3} term of (40); and that the third-order residual of $\log \Phi_{n,k}(v)$ times N^4 is nearly constant at $n = 60, 120, 240$ for $k = 2, \dots, 5$ and several v . A naive principal logarithm failed at $k = 5$, $v = i$, which shows that the branch convention (67) is needed. These checks do not replace an independent review of the proofs.

II.A Direct verification of the weighted gamma tail

This appendix is unnecessary for Theorem II.4.2, but verifies the tail assertion in the original approach of Theorem 6.2. Let $T_N \sim \text{Gamma}(N + 1, 1)$, with density $t^N e^{-t}/N!$. The exact representation (27) is

$$\Phi_{n,k}(v) = \mathbb{E} R_k(v/T_N)^n.$$

For $E_N = \{t : |t - N| > N^{2/3}\}$, positivity and finite expansion give

$$\mathbb{E}[|R_k(v/T_N)|^n \mathbf{1}_{E_N}(T_N)] \leq \sum_{M=0}^d c_{n,M} V^M \mathbb{P}\{\text{Gamma}(N - M + 1, 1) \in E_N\}, \quad |v| \leq V. \quad (72)$$

Multiplying the original density by t^{-M} changes its shape to $N - M + 1$, with normalizing factor $(N - M)!/N!$, which proves (72). All shapes are positive.

Split at $M_N = \lfloor N^{1/3} \rfloor$. For $M \leq M_N$, the new mean $q = N - M + 1$ is within $M_N + 1$ of N . For all large N , membership in E_N implies deviation from q by more than $N^{2/3}/2$. The gamma moment-generating function $(1 - s)^{-q}$ and Chernoff optimization give

$$\mathbb{P}(T - q \geq a) \leq e^{-a^2/[2(q+a)]}, \quad \mathbb{P}(T - q \leq -a) \leq e^{-a^2/(2q)} \quad (0 < a < q).$$

For the upper bound optimize at $s = a/(q + a)$; for the lower, at $s = -a/(q - a)$. The resulting rate functions dominate the stated quadratic expressions by integrating their derivatives. Thus the $M \leq M_N$ contribution in (72) is at most $2e^{\lambda V - cN^{1/3}}$ for an absolute $c > 0$ and sufficiently large N .

The other terms total at most $\sum_{M>M_N}(\lambda V)^M/M!$. For fixed $a = \lambda V > 0$, once $M_N + 2 \geq 2a$ this is bounded by

$$\frac{2a^{M_N+1}}{(M_N+1)!} \leq 2 \left(\frac{ea}{M_N+1} \right)^{M_N+1} = \exp\{-\Omega(N^{1/3} \log N)\}.$$

For $V = 0$ it vanishes. The entire weighted complement is therefore smaller than every power of N^{-1} , uniformly on the disk. The same factorial bound for degrees $M \geq \varepsilon N$, fixed $\varepsilon > 0$, gives $\exp\{-\varepsilon N \log N + O(N)\}$ (or zero if beyond support). No uncontrolled prefactor C^n is introduced.

References

- [1] OEIS Foundation Inc., *A330266: Number of ways to shuffle a deck of $4n$ cards, with four cards in each of n ranks, so that adjacent cards have different ranks*, <https://oeis.org/A330266>, accessed 1 October 2026.
- [2] OEIS Foundation Inc., *A007060: Number of ways n married couples can sit in a row without any spouses next to each other*, <https://oeis.org/A007060>, accessed 1 October 2026.
- [3] OEIS Foundation Inc., *A193624: Number of ways n triples can sit in a row without any siblings next to each other*, <https://oeis.org/A193624>, accessed 1 October 2026.
- [4] OEIS Foundation Inc., *A114938: Permutations of $\{1, 1, 2, 2, \dots, n, n\}$ with no equal consecutive terms*, <https://oeis.org/A114938>, accessed 1 October 2026.
- [5] OEIS Foundation Inc., *A193638: Permutations of the balanced multiplicity-three multiset with no equal consecutive terms*, <https://oeis.org/A193638>, accessed 1 October 2026.
- [6] OEIS Foundation Inc., *A321633: Permutations of the balanced multiplicity-four multiset with no equal consecutive terms*, <https://oeis.org/A321633>, accessed 1 October 2026.
- [7] H. Eriksson and A. Martin, *Enumeration of Carlitz multipermutations*, arXiv:1702.04177, 2017.
- [8] J. Enouen, *What is the Perfect Shuffle?*, arXiv:1911.07426, 2019.
- [9] E.-M. Mehiri, *Bijections Between Smirnov Words and Hamiltonian Cycles in Complete Multipartite Graphs*, arXiv:2510.26597v2, 2026.
- [10] [Added in the batch-77P5 write, from Part II.] I. M. Gessel, *Generalized rook polynomials and orthogonal polynomials*, in *q-Series and Partitions*, IMA Vol. 18, Springer, 1989, pp. 159–176. Author-hosted manuscript revised 30 September 1988: <https://people.brandeis.edu/~gessel/homepage/papers/rookp.pdf>.
- [11] [Added in the batch-77P5 write, from Part II.] J. Taylor, *Counting words with Laguerre series*, Electronic Journal of Combinatorics 21(2), P2.1, 2014. <https://www.combinatorics.org/ojs/index.php/eljc/article/download/v21i2p1/pdf/>.
- [12] R. P. Stanley, *Enumerative Combinatorics, Volume 1*, second edition, Cambridge University Press, 2012.
- [13] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.

- [14] NIST Digital Library of Mathematical Functions, chapters on the gamma and Laguerre functions, <https://dlmf.nist.gov/>, accessed 1 October 2026.
- [15] [Added in the batch-73O2 write.] V. Reshetnikov and contributors, *Transseries and Inversion*, canonical volume of the ProveIt series-and-transseries collection. Its source is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`. Cited here: Part “Inversion: the apparatus for a rapidly growing function” (labels `p0:thm:perturbed-inversion`, `p0:thm:staircase`; Part II also names `p0:thm:lambert-core`), chapter on the increasing inverse of Gamma (`p6:thm:gamma`) and chapter “The subfactorial” (`p8:sec:top`). Repository state of placement commit `6e193dd4f`.